$$f(n,m)=\sum_{i=0}^{n}\sum_{j=0}^{m}(-1)^{i+j}(n,i)(m,j)g(i,j)\iff g(n,m)=\sum_{i=0}^{n}\sum_{j=0}^{m}(-1)^{i+j}(n,i)(m,j)f(i,j)$$